

The Z-Transform Notes (By Ankit K.)

1. Definition of Z-Transform

The Z-transform converts a discrete-time signal, which is a sequence of real or complex numbers, into a complex frequency-domain representation.

- **Bilateral (Two-Sided) Z-Transform:**

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

- **Unilateral (One-Sided) Z-Transform** (used when signals are zero for $n < 0$):

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

- **Region of Convergence (ROC):** The set of points in the complex z -plane for which the Z-transform summation converges (i.e., $|X(z)| < \infty$).

Numerical 1: Finite Sequence

Problem: Find the Z-transform of the sequence $x(n) = \{2, \mathbf{4}, 5, 7\}$, where the bold number indicates the value at $n = 0$.

Solution:

The sequence values are $x(-1) = 2$, $x(0) = 4$, $x(1) = 5$, $x(2) = 7$.

Using the defining formula:

$$X(z) = \sum_{n=-1}^2 x(n)z^{-n} = 2z^1 + 4z^0 + 5z^{-1} + 7z^{-2}$$

$$X(z) = 2z + 4 + \frac{5}{z} + \frac{7}{z^2}$$

ROC: Entire z -plane except $z = 0$ and $z = \infty$.

Numerical 2: Right-Sided Exponential Sequence

Problem: Determine the Z-transform and ROC of $x(n) = (0.5)^n u(n)$.

Solution:

Since $u(n) = 1$ for $n \geq 0$ and 0 for $n < 0$, the limits of summation become 0 to ∞ :

$$X(z) = \sum_{n=0}^{\infty} (0.5)^n z^{-n} = \sum_{n=0}^{\infty} (0.5z^{-1})^n$$

This is an infinite geometric series of the form $\sum a^n = \frac{1}{1-a}$, provided $|a| < 1$.

$$X(z) = \frac{1}{1 - 0.5z^{-1}} = \frac{z}{z - 0.5}$$

ROC: $|0.5z^{-1}| < 1 \implies |z| > 0.5$.

2. Properties of Z-Transform

Note: Let $Z\{x(n)\} = X(z)$ and $Z\{y(n)\} = Y(z)$.

A. Linearity

Property: $Z\{a \cdot x(n) + b \cdot y(n)\} = aX(z) + bY(z)$

Numerical: Find the Z-transform of $x(n) = 3(2^n)u(n) - 4(0.5^n)u(n)$.

Solution:

Using the standard pair $Z\{a^n u(n)\} = \frac{z}{z-a}$:

$$X(z) = 3 \left(\frac{z}{z-2} \right) - 4 \left(\frac{z}{z-0.5} \right)$$

ROC: Intersection of $|z| > 2$ and $|z| > 0.5$, which is $|z| > 2$.

B. Time Shifting

Property: $Z\{x(n-k)\} = z^{-k}X(z)$

Numerical: Find the Z-transform of $y(n) = u(n-3)$.

Solution:

We know $Z\{u(n)\} = \frac{z}{z-1}$ with ROC $|z| > 1$.

Using the time-shifting property with $k = 3$:

$$Y(z) = z^{-3} \left(\frac{z}{z-1} \right) = \frac{1}{z^2(z-1)}$$

ROC: $|z| > 1$.

C. Scaling in Z-Domain

Property: $Z\{a^n x(n)\} = X(a^{-1}z) = X\left(\frac{z}{a}\right)$

Numerical: Given $Z\{u(n)\} = \frac{z}{z-1}$, find the Z-transform of $y(n) = (0.2)^n u(n)$ using the scaling property.

Solution:

Here, $x(n) = u(n)$ and $a = 0.2$.

$$Y(z) = X\left(\frac{z}{0.2}\right) = \frac{\frac{z}{0.2}}{\frac{z}{0.2} - 1} = \frac{z}{z - 0.2}$$

ROC: $|z| > 0.2$.

D. Time Reversal

Property: $Z\{x(-n)\} = X(z^{-1})$

Numerical: Find the Z-transform of $y(n) = u(-n)$.

Solution:

We know $Z\{u(n)\} = \frac{1}{1-z^{-1}}$. Substitute z with z^{-1} :

$$Y(z) = \frac{1}{1 - (z^{-1})^{-1}} = \frac{1}{1 - z}$$

ROC: $|z^{-1}| > 1 \implies |z| < 1$.

E. Differentiation in Z-Domain

Property: $Z\{n \cdot x(n)\} = -z \frac{dX(z)}{dz}$

Numerical: Find the Z-transform of $y(n) = nu(n)$.

Solution:

Let $x(n) = u(n)$, so $X(z) = \frac{z}{z-1}$.

$$Y(z) = -z \frac{d}{dz} \left(\frac{z}{z-1} \right) = -z \left[\frac{(z-1)(1) - z(1)}{(z-1)^2} \right] = \frac{z}{(z-1)^2}$$

ROC: $|z| > 1$.

3. Z-Transforms of Standard Signals

- **Unit Impulse** $\delta(n)$: $X(z) = 1$, ROC: All z
- **Unit Step** $u(n)$: $X(z) = \frac{z}{z-1}$, ROC: $|z| > 1$
- **Exponential** $a^n u(n)$: $X(z) = \frac{z}{z-a}$, ROC: $|z| > |a|$
- **Ramp** $nu(n)$: $X(z) = \frac{z}{(z-1)^2}$, ROC: $|z| > 1$

Numerical 1: Combination of Standard Signals

Problem: Find the Z-transform of $x(n) = 5\delta(n) + 2\left(\frac{1}{3}\right)^n u(n)$.

Solution:

Look up the standard transforms from the table:

$$X(z) = 5(1) + 2 \left(\frac{z}{z - 1/3} \right) = 5 + \frac{2z}{z - 1/3}$$

ROC: $|z| > 1/3$.

Numerical 2: Delayed Standard Signal

Problem: Find the Z-transform of $x(n) = \delta(n - 2) + 4u(n - 1)$.

Solution:

Using standard transforms and the time-shifting property:

$$X(z) = z^{-2}(1) + 4z^{-1} \left(\frac{z}{z-1} \right) = \frac{1}{z^2} + \frac{4}{z-1}$$

ROC: $|z| > 1$.

4. Inverse Z-Transform

The inverse Z-transform converts the frequency-domain signal $X(z)$ back into the discrete-time domain signal $x(n)$.

Numerical 1: Partial Fraction Expansion

Problem: Find the causal inverse Z-transform of $X(z) = \frac{z}{(z-1)(z-2)}$.

Solution:

First, divide $X(z)$ by z :

$$\frac{X(z)}{z} = \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

Solve for constants:

$$A = \left. \frac{1}{z-2} \right|_{z=1} = -1$$

$$B = \left. \frac{1}{z-1} \right|_{z=2} = 1$$

Re-multiply by z :

$$X(z) = -\frac{z}{z-1} + \frac{z}{z-2}$$

For a causal sequence (ROC $|z| > 2$), the inverse is:

$$x(n) = [-(1)^n + 2^n] u(n) = (2^n - 1)u(n)$$

Numerical 2: Partial Fraction Expansion (Distinct Roots)

Problem: Find the causal inverse Z-transform of $X(z) = \frac{z}{(z-3)(z-4)}$.

Solution:

Divide $X(z)$ by z :

$$\frac{X(z)}{z} = \frac{1}{(z-3)(z-4)} = \frac{A}{z-3} + \frac{B}{z-4}$$

Solve for constants:

$$A = \left. \frac{1}{z-4} \right|_{z=3} = \frac{1}{-1} = -1$$

$$B = \frac{1}{z-3} \Big|_{z=4} = \frac{1}{1} = 1$$

Re-multiply by z :

$$X(z) = -\frac{z}{z-3} + \frac{z}{z-4}$$

For a causal sequence (ROC $|z| > 4$):

$$x(n) = [-3^n + 4^n] u(n)$$

5. Analysis of Systems using Z-Transform

Numerical 1: Finding System Output (Zero Initial Conditions)

Problem: For the causal LTI system with transfer function $H(z) = \frac{z}{z-0.8}$, find the step response $y(n)$.

Solution:

The input is a unit step $x(n) = u(n)$, so $X(z) = \frac{z}{z-1}$.

$$Y(z) = H(z)X(z) = \left(\frac{z}{z-0.8} \right) \left(\frac{z}{z-1} \right)$$

Divide by z to perform partial fractions:

$$\frac{Y(z)}{z} = \frac{z}{(z-0.8)(z-1)} = \frac{A}{z-0.8} + \frac{B}{z-1}$$

$$A = \frac{z}{z-1} \Big|_{z=0.8} = \frac{0.8}{-0.2} = -4$$

$$B = \frac{z}{z-0.8} \Big|_{z=1} = \frac{1}{0.2} = 5$$

$$Y(z) = -4 \frac{z}{z-0.8} + 5 \frac{z}{z-1}$$

Taking the inverse Z-transform:

$$y(n) = [-4(0.8)^n + 5(1)^n] u(n) = [5 - 4(0.8)^n] u(n)$$

Numerical 2: System Analysis with Initial Conditions

Problem: Determine the total response $y(n)$ for $n \geq 0$ of a discrete-time system described by the difference equation:

$$y(n) - 0.5y(n-1) = x(n)$$

Given the input $x(n) = (0.2)^n u(n)$ and the initial condition $y(-1) = 2$.

Solution:

Applying the unilateral Z-transform property for time delay with initial conditions ($Z\{y(n-1)\} = z^{-1}Y(z) + y(-1)$):

$$Y(z) - 0.5[z^{-1}Y(z) + y(-1)] = X(z)$$

Substitute $X(z) = \frac{1}{1-0.2z^{-1}}$ and $y(-1) = 2$:

$$Y(z)(1 - 0.5z^{-1}) - 0.5(2) = \frac{1}{1 - 0.2z^{-1}}$$

$$Y(z)(1 - 0.5z^{-1}) = 1 + \frac{1}{1 - 0.2z^{-1}}$$

$$Y(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

Partial fraction for the second term (let $v = z^{-1}$):

$$\frac{1}{(1 - 0.5v)(1 - 0.2v)} = \frac{A}{1 - 0.5v} + \frac{B}{1 - 0.2v}$$

$$A = \left. \frac{1}{1 - 0.2v} \right|_{v=2} = \frac{1}{1 - 0.4} = \frac{5}{3}$$

$$B = \left. \frac{1}{1 - 0.5v} \right|_{v=5} = \frac{1}{1 - 2.5} = -\frac{2}{3}$$

Combining the terms into total $Y(z)$:

$$Y(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{5/3}{1 - 0.5z^{-1}} - \frac{2/3}{1 - 0.2z^{-1}} = \frac{8/3}{1 - 0.5z^{-1}} - \frac{2/3}{1 - 0.2z^{-1}}$$

Taking the inverse Z-transform gives the total response:

$$y(n) = \left[\frac{8}{3}(0.5)^n - \frac{2}{3}(0.2)^n \right] u(n)$$

6. Convolution Sum using Z-Transform

The convolution property states that convolution in the time domain corresponds to multiplication in the Z-domain:

$$x_1(n) * x_2(n) \longleftrightarrow X_1(z) \cdot X_2(z)$$

Numerical 1: Convolution of Finite Sequences

Problem: Find the convolution of $x(n) = \{1, 2, 1\}$ (origin at first term) and $h(n) = \{1, -1, 1\}$ using the Z-transform.

Solution:

First, find the Z-transforms:

$$X(z) = 1 + 2z^{-1} + z^{-2}$$

$$H(z) = 1 - z^{-1} + z^{-2}$$

Multiply $Y(z) = X(z)H(z)$:

$$Y(z) = (1 + 2z^{-1} + z^{-2})(1 - z^{-1} + z^{-2})$$

Expanding the terms:

$$Y(z) = 1(1 - z^{-1} + z^{-2}) + 2z^{-1}(1 - z^{-1} + z^{-2}) + z^{-2}(1 - z^{-1} + z^{-2})$$

$$Y(z) = 1 - z^{-1} + z^{-2} + 2z^{-1} - 2z^{-2} + 2z^{-3} + z^{-2} - z^{-3} + z^{-4}$$

$$Y(z) = 1 + z^{-1} + 0z^{-2} + z^{-3} + z^{-4}$$

Taking the inverse Z-transform yields the sequence:

$$y(n) = \{1, 1, 0, 1, 1\}$$

Numerical 2: Convolution of Infinite Sequences

Problem: Convolve $x(n) = (0.5)^n u(n)$ with $h(n) = (0.2)^n u(n)$ using the Z-transform.

Solution:

Find individual Z-transforms:

$$X(z) = \frac{z}{z - 0.5}, \quad H(z) = \frac{z}{z - 0.2}$$

Multiply them to find $Y(z)$:

$$Y(z) = \frac{z^2}{(z - 0.5)(z - 0.2)}$$

Divide by z :

$$\frac{Y(z)}{z} = \frac{z}{(z - 0.5)(z - 0.2)} = \frac{A}{z - 0.5} + \frac{B}{z - 0.2}$$

$$A = \left. \frac{z}{z - 0.2} \right|_{z=0.5} = \frac{0.5}{0.3} = \frac{5}{3}$$

$$B = \left. \frac{z}{z - 0.5} \right|_{z=0.2} = \frac{0.2}{-0.3} = -\frac{2}{3}$$

$$Y(z) = \frac{5}{3} \frac{z}{z - 0.5} - \frac{2}{3} \frac{z}{z - 0.2}$$

Taking the inverse Z-transform:

$$y(n) = \left[\frac{5}{3}(0.5)^n - \frac{2}{3}(0.2)^n \right] u(n)$$

Inverse Z-Transform for All Possible ROCs

Problem:

Find the inverse Z-transform of the following function:

$$X(z) = \frac{z}{(z - 0.5)(z - 2)}$$

Determine the sequence $x(n)$ for all three possible Regions of Convergence (ROCs):

1. $|z| > 2$
2. $0.5 < |z| < 2$
3. $|z| < 0.5$

Step 1: Partial Fraction Expansion

Before applying the ROC conditions, we must expand $X(z)$ into standard partial fraction forms. To avoid fractional z terms in the numerator, we first divide $X(z)$ by z :

$$\frac{X(z)}{z} = \frac{1}{(z - 0.5)(z - 2)}$$

Now, perform the partial fraction expansion:

$$\frac{1}{(z - 0.5)(z - 2)} = \frac{A}{z - 0.5} + \frac{B}{z - 2}$$

Solving for A and B:

- For A : Multiply by $(z - 0.5)$ and set $z = 0.5$

$$A = \left. \frac{1}{z - 2} \right|_{z=0.5} = \frac{1}{0.5 - 2} = \frac{1}{-1.5} = -\frac{2}{3}$$

- For B : Multiply by $(z - 2)$ and set $z = 2$

$$B = \left. \frac{1}{z - 0.5} \right|_{z=2} = \frac{1}{2 - 0.5} = \frac{1}{1.5} = \frac{2}{3}$$

Substitute A and B back into the equation and multiply by z :

$$X(z) = -\frac{2}{3} \left(\frac{z}{z - 0.5} \right) + \frac{2}{3} \left(\frac{z}{z - 2} \right)$$

Standard transform pairs based on the ROC:

- **Right-sided sequence (causal):** $Z^{-1} \left\{ \frac{z}{z-a} \right\} = a^n u(n)$ for $|z| > |a|$
- **Left-sided sequence (anti-causal):** $Z^{-1} \left\{ \frac{z}{z-a} \right\} = -a^n u(-n - 1)$ for $|z| < |a|$

Step 2: Evaluating for Each ROC

Case 1: ROC is $|z| > 2$

Because $|z| > 2$, it is inherently true that $|z| > 0.5$. The ROC is entirely outside the outermost pole. This indicates that **both terms** correspond to right-sided (causal) sequences.

Applying the standard right-sided inverse formula to both terms:

$$x_1(n) = -\frac{2}{3}(0.5)^n u(n) + \frac{2}{3}(2)^n u(n)$$

$$\mathbf{x}_1(\mathbf{n}) = \frac{2}{3} [2^n - (0.5)^n] \mathbf{u}(\mathbf{n})$$

Case 2: ROC is $0.5 < |z| < 2$

This ROC is an annular ring between the two poles. This implies a two-sided sequence.

- For the pole at 0.5, the ROC is $|z| > 0.5$, meaning this term produces a **right-sided** sequence.
- For the pole at 2, the ROC is $|z| < 2$, meaning this term produces a **left-sided** sequence.

Applying the respective inverse formulas:

$$x_2(n) = -\frac{2}{3}(0.5)^n u(n) + \frac{2}{3} [-(2)^n u(-n-1)]$$

$$\mathbf{x}_2(\mathbf{n}) = -\frac{2}{3}(0.5)^n \mathbf{u}(\mathbf{n}) - \frac{2}{3}(2)^n \mathbf{u}(-\mathbf{n}-1)$$

Case 3: ROC is $|z| < 0.5$

Because $|z| < 0.5$, it is inherently true that $|z| < 2$. The ROC is entirely inside the innermost pole. This indicates that **both terms** correspond to left-sided (anti-causal) sequences.

Applying the standard left-sided inverse formula to both terms:

$$x_3(n) = -\frac{2}{3} [-(0.5)^n u(-n-1)] + \frac{2}{3} [-(2)^n u(-n-1)]$$

$$\mathbf{x}_3(\mathbf{n}) = \frac{2}{3} [(0.5)^n - 2^n] \mathbf{u}(-\mathbf{n}-1)$$